

Bhavdeep Arora

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Education

Toronto Metropolitan University

Bachelor of Science (Honours) Computer Science — Software Engineering Concentration

Toronto, ON

July 2026

Experiences

Data for Good Toronto | Data Analyst

May 2026 – Present

- Evaluated and cleaned **15+** datasets across **6** non-profit partner organizations, delivering structured analysis that informed program resource allocation decisions for community-focused initiatives
- Built data-cleaning and transformation workflows in Python and pandas across **end-to-end** analytics pipelines, reducing manual data-prep time by **35%** and enabling faster turnaround on partner-facing reports

SciXchange | Technical Assistant

September 2023 – April 2024

- Guided **500+** visitors through **12** interactive science exhibits per shift, improving average visitor engagement time by **20%** through clear translation of technical STEM concepts, while coordinating with a team of **8** volunteers and staff under time-sensitive conditions

Technical Projects

🌐 bhavdeeparora.dev/projects

Thunderhead — Reverse Proxy for Automated Bot Detection | Go, HTTP, Behavioral Analysis | 🌐 Live | 📄 Github

- Engineered a reverse proxy in Go analyzing inbound HTTP traffic across **five behavioral heuristics** — crawl velocity, robots.txt violations, request frequency, and header anomalies — enabling bot detection with **zero** third-party security dependencies
- Developed a configurable **risk-scoring engine** that assigns request scores from 0–100 across **12** weighted signals and applies graduated mitigations (allow, tarpit, or block), correctly flagging **94% of automated traffic across 500,000 requests** while maintaining a **2% false-positive rate** on legitimate users

Groat — Self-Hosted LLM Cost Optimization Proxy | Rust, Axum, SQLite, LanceDB | 🌐 Live | 📄 Github

- Architected a self-hosted, OpenAI-compatible LLM proxy in Rust as a **4-crate workspace** (server, core logic, provider layer, storage), enabling drop-in cost optimization with **zero** application code changes
- Implemented **semantic response caching** via bge-small-en-v1.5 embeddings and LanceDB similarity search (0.95 threshold), automatic prompt-cache-boundary injection to claim provider-side **50–90% cached-token discounts**, and user-defined cost-tier routing, cutting redundant API spend by **31%** across **8,200+** logged requests

Funes — Local-First Semantic Memory System | Rust, Ollama, SQLite, CLI | 🌐 Live | 📄 Github

- Indexed **4000+ files** across **1500 sources** (notes, shell history, documents) at **285 files/sec**, by building a local-first semantic memory system in Rust requiring **zero cloud infrastructure**
- Achieved **85ms average query latency** and **91% top-5 retrieval accuracy** across **3,400 indexed documents**, by architecting a background indexing daemon with filesystem monitoring, chunking pipelines, and SQLite-backed vector storage

Verrere — Swipe-Based Book Discovery Platform | Next.js 15, TypeScript, Prisma, PostgreSQL | 🌐 Live | 📄 Github

- Achieved a **68%** swipe-to-save rate across **1,200+ active users**, by building a full-stack book discovery platform with a swipe-based recommendation interface, **integrating the Hardcover GraphQL API with PostgreSQL/Prisma persistence**

Research & Publications

🌐 bhavdeeparora.dev/research

Precursor — Cross-Asset Momentum Spillover Research | Python, statsmodels, VAR, NetworkX | 📄 SSRN Paper | 📄 Github

- Tested **Granger causality** between commodity futures and sector equity ETF momentum (2010–2026, **five** commodity-equity pairs), confirming **all five** hypothesised spillovers at $\alpha = 0.05$ (Gold→GDX: $F=18.90$, $p<0.001$)
- Ran rolling Granger tests and Chow structural break analysis to show spillovers are **regime-dependent** rather than persistent, then validated with a signal-based backtest achieving a **Sharpe ratio of 1.4**

Lacunae — Deep Learning MRI Reconstruction | Python, PyTorch, U-Net, fastMRI | 📄 Github

- Trained a **7.7M-parameter U-Net** on ~ 100 GB of fastMRI single-coil knee MRI data to reconstruct diagnostic-quality images from **8× undersampled** k-space
- Achieved **SSIM: 0.7803** and **PSNR: 30.8 dB** on held-out validation scans, an **18% improvement in SSIM** over zero-filled baseline reconstruction

Technical Skills

Languages: Python, Go, Java, JavaScript/TypeScript, Bash, Fish, Nim, HTML, CSS/SCSS

DevOps Tools: Linux, AWS, Cloudflare, Docker (Swarm, Compose, BuildX), Caddy, Nginx, Traefik, WireGuard, Tailscale, GitHub Actions

Tools & Frameworks: Django, FastAPI, Flask, React, Svelte, Astro, Gin, Celery, Postman, Tailscale, Git

Databases: PostgreSQL (PostGIS, PgVector), SQLite, Redis, MySQL, Snowflake

Certifications

AWS Certified Cloud Practitioner | Amazon Web Services

August 2026

- Built a foundation in cloud architecture, security, networking, and cost-efficient deployment strategies across core AWS services

Google Data Analytics Professional Certificate | Google

August 2026

Google Cloud Professional Data Engineer | Google Cloud (In Progress)

September 2026